

Linear Algebra Preliminary Exam

August 18, 2025

1. Let $T : V \rightarrow V$ be a linear map where V is a linear vector space of dimension $n \geq 2$ over $\mathbb{R}$, and suppose that for some vector $v \in V$, the n vectors $v, Tv, \dots, T^{n-1}v$ are linearly independent. Prove the minimal polynomial of T has degree n .
2. Let A be an $n \times n$ real symmetric matrix. Show that A has at least one real eigenvalue λ , such that

$$\lambda \geq \max_{1 \leq k \leq n} a_{kk}.$$

3. Let V be a non-trivial space of continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $f \in V$, we have either $f(x) \geq 0$ for all $x \in \mathbb{R}$ or $f(x) \leq 0$ for all $x \in \mathbb{R}$. Show that V is one-dimensional.
4. Let A be an $n \times n$ complex normal matrix. Its operator norm $\|A\|$ is defined by

$$\|A\| = \sup_{v \in \mathbb{C}^n, v \neq 0} \frac{|Av|}{|v|}$$

where $|\cdot|$ represents the Euclidean norm of vectors in $\mathbb{C}^n$. Show that if $\|A\| = 1$ and its trace $\text{tr } A = n$, then A is the identity matrix.

5. Let P be an $n \times n$ complex matrix such that $P^n = 0$ but $P^{n-1} \neq 0$. Let M_n be the linear space consists of all $n \times n$ complex matrices. Consider the linear map $L : M_n \rightarrow M_n$ defined by

$$L : S \mapsto SP \text{ for any } S \in M_n$$

Find the rank of L and justify your answer.

6. Let V be a finite-dimensional complex vector space, $A : V \rightarrow V$ linear. Prove that the following are equivalent:
 - a. There exists an inner product on V for which A is self-adjoint.
 - b. A is diagonalizable and all eigenvalues are real.